

DIY Mini Knitting Machine / Loop Loom Starter Guide

This guide is a practical step-by-step plan for building a small 4–6 needle knitting prototype using latch needles, then upgrading it to a crank and later a motorized Arduino-driven system.

1. Recommended First Prototype Dimensions

Outer disk diameter : 70 mm
Center hole : 25 mm
Needle circle : 32 mm
Needle slot width : 4.5 mm
Needle slot depth : 3 mm
Needle slot length : 28 mm

Top View Sketch

N1
N4 N2
N3
0 = center hole

2. Build Order

Phase 1 - One needle prototype
Phase 2 - Four needle ring
Phase 3 - Yarn guide
Phase 4 - Hanging weight
Phase 5 - Springs
Phase 6 - Cam ring
Phase 7 - Hand crank
Phase 8 - Stepper motor + Arduino

3. Improved Non-Twisting Design

Instead of rotating the knitted tube, keep the fabric hanging straight down and move only the yarn feeder.

Rotating Yarn Feeder
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N1 N2
N4 N3

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Fabric Tube
(stationary)

4. Side View Concept

Yarn Guide Arm
|
V

Needle Holder Ring

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||
Knitted Tube
||
Weight

5. Shopping List

Already owned:

- 6 latch needles

Buy first:

- Plywood or MDF
- Acrylic yarn
- Small screws
- Sandpaper
- Pen springs
- Washers/sinker weight

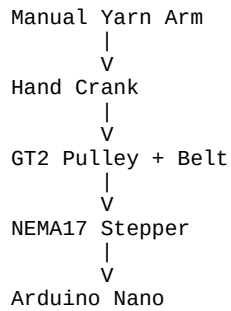
Buy later:

- Bearings (608)
- M6 rod
- GT2 belt
- GT2 pulley
- NEMA17 stepper
- A4988 driver
- Arduino Nano

6. Stitch Formation Sequence

1. Old loop sits on needle
2. Needle rises
3. Latch opens
4. Yarn enters hook
5. Needle lowers
6. Old loop closes latch
7. Old loop falls off
8. New stitch remains

7. Motor Upgrade Path



Goal

Get one needle knitting consistently before adding more needles, cams, or electronics. Once a single needle works reliably, replicate the mechanism around the ring.